Large Language Models Can Self-Improve in Long-context Reasoning

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Abstract

Large language models (LLMs) have achieved substantial progress in processing long contexts but still struggle with long-context reasoning. Existing approaches typically involve fine-tuning LLMs with synthetic data, which depends on annotations from human experts or advanced models like GPT-4, thus restricting further advancements. To address this issue, we investigate the potential for LLMs to self-improve in long-context reasoning and propose SEALONG, an approach specifically designed for this purpose. This approach is straightforward: we sample multiple outputs for each question, score them with Minimum Bayes Risk, and then apply supervised fine-tuning or preference optimization based on these outputs. Extensive experiments on several leading LLMs demonstrate the effectiveness of SEALONG, with an absolute improvement of 4.2 points for Llama-3.1-8B-Instruct. Furthermore, SEALONG achieves superior performance compared to prior approaches that depend on data produced by human experts or advanced models. We anticipate that this work will open new avenues for self-improvement techniques in long-context scenarios, which are essential for the continual advancement of LLMs.[1]

1 Introduction

Large language models (LLMs) with long-context processing capabilities have spurred a range of novel applications, such as repository-level coding assistance (Jimenez et al., 2024), multi-document analysis (Wang et al., 2024b) and autonomous agents (Ma et al., 2024). Delivering high-quality service in these domains requires LLMs to *reason effectively over long contexts*, necessitating the retrieval of essential details and integration of dispersed information throughout the reasoning process. While recent advancements have enabled LLMs to attain near-perfect accuracy on the needle-in-a-haystack (NIAH; Kamradt (2023); Li et al. (2024b)) task (Hsieh et al., 2024; Yen et al., 2024b; Dubey et al., 2024), which involves locating evidence within vast amounts of irrelevant text, substantial performance declines persist on tasks that require reasoning over long contexts (Levy et al., 2024; Hsieh et al., 2024; Vodrahalli et al., 2024; Yen et al., 2024b; Li et al., 2024a), limiting their applicability in real-word scenarios.

To address this limitation, recent studies have investigated fine-tuning LLMs to improve long-context reasoning, with effective data synthesis as a primary challenge. Two main approaches have emerged: one relies on human annotations (Chen et al., 2024b; Li et al., 2024c,a), which are expensive and difficult to scale; the other leverages expert models, such as GPT-4o (Hurst et al., 2024), for data synthesis. For example, Bai et al. (2024); Zhang et al. (2024c,b) apply self-instruct techniques (Wang et al., 2023b) to create long-context instruction-following data. Despite substantial progress, the dependence on pre-existing expert models limits the potential for achieving more advanced capabilities.

This work investigates whether LLMs can *self-improve in long-context reasoning*. Drawing on evidence of LLMs' near-perfect long-context retrieval and strong reasoning abilities in general domains (Bubeck et al., 2023; Zhong et al., 2024), we hypothesize that LLMs have untapped potential in long-context reasoning. Our preliminary studies show that refined prompting strategies achieve notable improvements over both default prompting methods and direct answer requests. Furthermore, scaling the number of sampled outputs reveals a marked performance gap between the optimal outputs and those derived via greedy search. These results suggest that LLMs hold substantial potential to advance in long-context reasoning.

Inspired by these observations, we propose a

[1]The repository can be accessed at https://github.com/SihengLi99/SEALONG.

Prompt	Llama-3.1-8B-Instruct			Llama-3.1-70B-Instruct		
	HotpotQA	MuSiQue	2WikiMQA	HotpotQA	MuSiQue	2WikiMQA
Default	55.5	33.0	66.0	60.0	54.0	77.0
Direct answer	49.0	28.5	55.0	61.5	51.5	74.0
Think step-by-step (Kojima et al., 2022)	62.5	**50.5**	77.5	75.5	62.5	85.0
Fact-and-reflection (Zhao et al., 2024b)	**67.0**	49.0	76.5	**78.0**	62.0	84.0
Plan-and-solve (Wang et al., 2023a)	64.0	49.5	**82.0**	74.0	**68.5**	**85.5**

Table 1: Comparison of various prompting methods. The best result is highlighted in bold.

Self-improving method for rEAsoning over LONG-contexts (SEALONG). This involves first sampling multiple reasoning trajectories from the LLM, then scoring each based on Minimum Bayes Risk (MBR) (Bickel and Doksum, 1977), which prioritizes outputs that are more consistent with others. This idea is intuitive, as reasoning trajectories that deviate from the majority are more likely to be hallucinations (Manakul et al., 2023; Farquhar et al., 2024). Following this, we can either conduct supervised fine-tuning using high-scoring outputs or apply preference optimization by utilizing both high-scoring and low-scoring outputs.

We apply SEALONG to several leading LLMs and conduct evaluations on multiple long-context reasoning tasks (Bai et al., 2023; Yang et al., 2018; Trivedi et al., 2022; Ho et al., 2020; Dasigi et al., 2021). The results reveal that LLMs can self-improve in long-context reasoning. Specifically, SEALONG raises the score of Llama-3.1-8B-Instruct (Dubey et al., 2024) from 50.8 to 55.0. Additionally, SEALONG enables Qwen-2.5-14B-Instruct (Yang et al., 2024a) to outperform its 32B variant (54.7 vs. 53.1). In comparison to previous synthetic data, SEALONG demonstrate notable improvement without requiring human or expert model annotation. We hope that SEALONG can pave the way for self-improving approaches in long-context scenarios, supporting the continual advancement of LLM capabilities.

2 Understanding the Potential of LLMs in Long-context Reasoning

We explore the potential of LLMs in long-context reasoning through experiments on three reasoning-intensive tasks from LongBench (Bai et al., 2023): HotpotQA (Yang et al., 2018), MuSiQue (Trivedi et al., 2022) and 2WikiMQA (Ho et al., 2020). These tasks involve handling multiple documents within the context and addressing multi-hop questions that span several paragraphs. Following previous work (Mallen et al., 2023; Asai et al., 2024; Yen et al., 2024b), we use substring exact match (SubEM) for evaluation, assessing whether the golden answer is included in the output.

2.1 Prompting Strategies Matter

Numerous long-context evaluation benchmarks assess LLMs by simply asking them to respond to a query based on a long context (Bai et al., 2023; An et al., 2024a; Zhang et al., 2024d; Wang et al., 2024b; Yen et al., 2024b). We suggest that this approach may underestimate LLMs' potential in long-context scenarios, particularly for questions requiring complex, multi-step reasoning to arrive at an answer. To further investigate this, we examine various prompting strategies for long-context reasoning, including:

- **Default**: Prompting the LLM with the long context and a question.
- **Direct Answer**: Asking the LLM to directly answer the question based on the long context.
- **Think Step-by-step**: Providing the LLM with the context, question, and an instruction to think step-by-step (Kojima et al., 2022).
- **Fact-and-reflection**: Providing the LLM with the long context, question, and an instruction to first identify the relevant information from the long context, and then carry out step-by-step reasoning and provide the answer (Zhao et al., 2024b; Li et al., 2024a).
- **Plan-and-solve**: Providing the LLM with the long context, question, and an instruction to first devise a plan and then follow it to solve the problem step-by-step (Wang et al., 2023a).

The detailed prompts for these strategies are presented in Tab. 9 (Appx. B). As shown in Tab. 1, prompting strategies play a crucial role in long-context reasoning. A notable performance gap

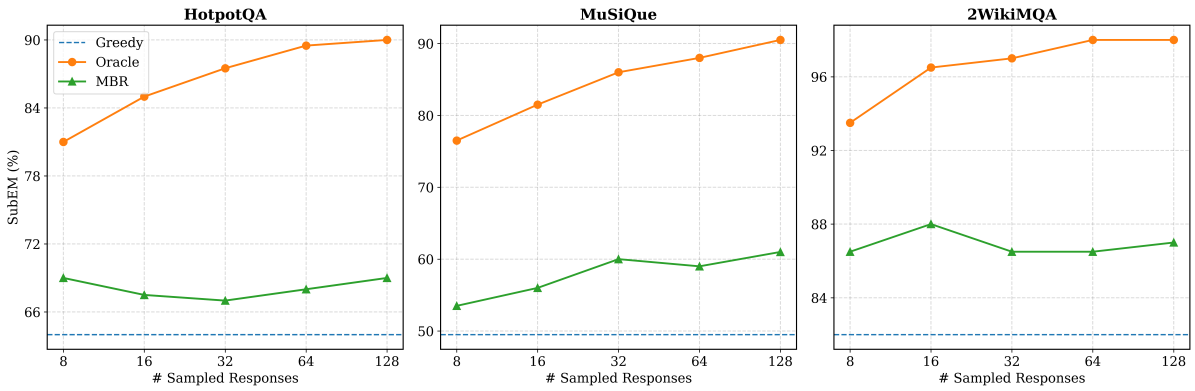

Figure 1: Scaling up the number of sampled outputs improves the performance of both the oracle sample and MBR decoding (§3.1). The results are based on Llama-3.1-8B-Instruct.

exists between default prompting and reasoning-targeted prompting strategies, aligning with observations in short-context tasks (Wei et al., 2022; Zhou et al., 2023). Manual inspection reveals that with an appropriate prompting strategy, the LLM breaks down multi-hop questions into simpler parts, addresses each part using the long context, and ultimately arrives at an answer.

2.2 The Potential of LLMs for Correct Long-context Reasoning

We further investigate the potential of LLMs for long-context reasoning by expanding the generation space. Specifically, we use temperature sampling to produce multiple outputs per question, evaluate each with SubEM, and designate the highest-scoring output as the oracle sample. As shown in Fig. 1, there is a notable gap between oracle performance and that of greedy search, even with just 8 outputs. Scaling up to 128 samples achieves over 90% correct answers. These results underscore the potential of LLMs for long-context reasoning and motivate the development of methods that enable LLMs to self-improve in this area.

3 SEALONG

Motivated by the potential of LLMs in long-context reasoning (§2), we propose SEALONG, a self-improving method for reasoning over long contexts. This approach consists of two stages: creating self-supervision and fine-tuning the model. An overview of SEALONG is provided in Fig. 2.

3.1 Self-supervision

We begin by leveraging plan-and-solve prompting (Wang et al., 2023a) to sample multiple reasoning trajectories for each question and its corresponding long context. The primary challenge lies in evaluating these outputs. The fundamental idea behind SEALONG is that correct reasoning trajectories typically exhibit higher semantic consistency. For example, they tend to follow similar planning steps and reference the same information within the long context. This observation aligns with hallucination detection methods (Manakul et al., 2023; Farquhar et al., 2024), where less consistent outputs are more likely to indicate hallucinations, representing incorrect reasoning in our scenario.

We formalize this idea using Minimum Bayes Risk (MBR) (Bickel and Doksum, 1977; Bertsch et al., 2023; Wu et al., 2024), which prioritizes outputs that exhibit higher consistency with others. In the MBR literature, the quality of an output is assessed by its expected utility under the model distribution (Bickel and Doksum, 1977; Kumar and Byrne, 2004; Tromble et al., 2008):

$$s(y) = \mathbb{E}_{y^* \sim \pi_\theta(y|x)} \left[u(y, y^*) \right]$$

Here, $s(y)$ is the score assigned to output y, where x denotes the input, including the long context, question and instruction. The term $\pi_\theta(y \mid x)$ represents the policy distribution of the LLM and the utility metric $u(y, y^*)$ assesses y based on y^*. We approximate this expectation using the Monte Carlo method with N sampled outputs:

$$s(y) \approx \frac{1}{N} \sum_{i=1}^{N} \left[u(y, y^*) \right]$$

The utility metric measures the consistency between two outputs. We use sentence embedding similarity as this metric, as it effectively captures

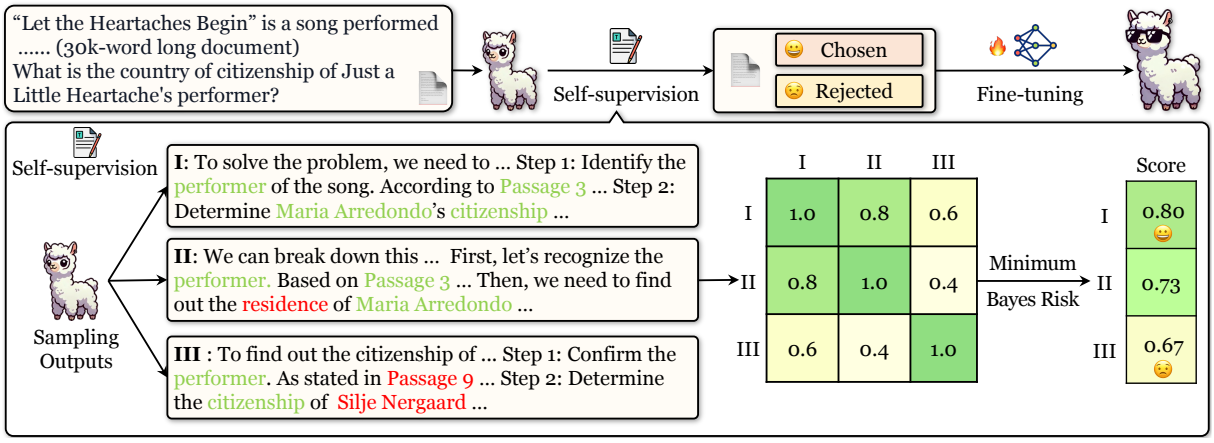

Figure 2: SEALONG consists of two stages: self-supervision creation and fine-tuning. Given a long context and a corresponding query, multiple outputs are sampled, each assigned a score based on Minimum Bayes Risk. Fine-tuning is then conducted using either the highest-scoring output for supervised fine-tuning or both high-scoring and low-scoring outputs for preference optimization.

the semantic alignment between the two reasoning trajectories. Formally:

$$u(y, y^*) = \text{Sim}\left(\text{Emb}(y), \text{Emb}(y^*)\right)$$

We employ a lightweight RoBERTa-based model (Liu, 2019) to embed outputs and measure similarity with the inner product. This approach allows us to assign each output y a score $s(y)$, and selecting the output with the highest score is referred to as MBR decoding (Bickel and Doksum, 1977; Bertsch et al., 2023; Wu et al., 2024). As demonstrated in Fig. 1, MBR decoding substantially surpasses greedy search: with absolute improvements of 11.5% on MuSiQue (Trivedi et al., 2022), 5.0% on HotpotQA (Yang et al., 2018), and 5.0% on 2WikiMultihopQA (Ho et al., 2020) when $N = 128$. These results highlight the potential for LLMs to self-improve by leveraging multiple samples and an effective evaluation metric based on output consensus, eliminating the need for human experts or advanced models. Furthermore, this evaluation approach produces preference pairs by contrasting high-scoring and low-scoring outputs, allowing straightforward preference optimization (Ouyang et al., 2022; Rafailov et al., 2024).

3.2 Fine-tuning

Leveraging self-provided supervision, we can either perform supervised fine-tuning on the highest-scoring outputs or apply preference optimization using preference pairs.

Supervised Fine-tuning. For supervised fine-tuning (SFT), we minimize the negative log-likelihood of the output as follows:

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{|y|} \log \pi_\theta(y \mid x)$$
$$= -\frac{1}{|y|} \sum_{i=1}^{|y|} \log \pi_\theta(y_i \mid x, y_{<i})$$

Here, y denotes the MBR decoding output.

Preference Optimization. Alternatively, we can conduct preference optimization to reinforce the tendency toward high-scoring outputs and reduce the likelihood of low-scoring outputs. Among the various preference optimization methods, we adopt the monolithic odds ratio preference optimization (ORPO) algorithm (Hong et al., 2024) due to its strong empirical performance. ORPO introduces an odds ratio loss to minimize the negative log odds ratio between a preferred output y_w and a less-preferred output y_l:

$$\mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \frac{\text{odds}_\theta(y_w|x)}{\text{odds}_\theta(y_l|x)} \right)$$

Here, σ represents the sigmoid function, and $\text{odds}_\theta(y|x)$ measures how much more likely y is to be generated than not:

$$\text{odds}_\theta(y|x) = \frac{\pi_\theta(y|x)}{1 - \pi_\theta(y|x)}$$

The final objective in ORPO combines SFT and OR losses, with a hyperparameter β controlling their relative importance:

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \beta \cdot \mathcal{L}_{\text{OR}}$$

Model	Qasper	MultiFieldQA-En	HotpotQA	MuSiQue	2WikiMQA	Avg.
Qwen-2.5-7B-Instruct (Yang et al., 2024a)	21.0	28.0	70.5	48.0	77.5	49.0
+ SEALONG	**26.0**	**29.3**	**72.5**	**51.5**	**79.5**	**51.8**
Qwen-2.5-14B-Instruct (Yang et al., 2024a)	21.0	**32.0**	73.0	52.0	83.0	52.2
+ SEALONG	**24.0**	30.0	**75.0**	**57.0**	**87.5**	**54.7**
Llama-3.1-8B-Instruct (Dubey et al., 2024)	29.0	29.3	64.0	49.5	82.0	50.8
+ SEALONG	**32.5**	**31.3**	**68.0**	**58.5**	**84.5**	**55.0**
Qwen-2.5-32B-Instruct (Yang et al., 2024a)	24.5	26.0	72.0	55.0	88.0	53.1
Qwen-2.5-72B-Instruct (Yang et al., 2024a)	27.0	28.7	**74.5**	58.5	**89.0**	55.5
Llama-3.1-70B-Instruct (Dubey et al., 2024)	**30.0**	**33.3**	74.0	**68.5**	85.5	**58.3**
GPT-4o (Hurst et al., 2024)	21.5	28.0	**74.5**	64.0	84.0	54.4

Table 2: Main evaluation results. Substring exact match (SubEM) serves as the evaluation metric, with the top-performing results emphasized in bold. SEALONG utilizes the training set of MuSiQue with self-supervision (§3.1), and its performance on other tasks demonstrates the generalization ability of SEALONG.

Task	# Example	Max Tokens	Avg. Tokens
Qasper	200	21,110	4,921
MultiFieldQA-en	150	14,947	6,888
HotpotQA	200	16,322	12,779
MuSiQue	200	16,335	15,542
2WikiMultihopQA	200	16,319	7,096

Table 3: Statistics of evaluation tasks, with token counts calculated using the tokenizer of Llama-3.1-8B-Instruct.

Model	Avg. Long-context	Avg. Output Tokens
Qwen-2.5-Instruct 7B	49.0	375
+ SEALONG	51.8	371
Llama-3.1-Instruct 8B	50.8	289
+ SEALONG	55.0	295

Table 4: Average performance on long-context tasks (Tab. 2) and average token count in model predictions for these tasks, measured with the model's tokenizer.

In our implementation, we use the MBR decoding output as y_w and randomly select a low-scoring output to serve as y_l.

4 Experiments

4.1 Implementation

SEALONG requires query and long-context pairs to synthesize training data. Specifically, we leverage the training dataset of MuSiQue (Trivedi et al., 2022), where each question is related to several Wikipedia documents. To achieve a specified number of tokens in the context, we randomly sample some unrelated documents, shuffle them with the related ones and concatenate them into a single context. We use the original questions in MuSiQue without the annotated answer, relying on the LLM to produce self-supervision (§3.1). For each question, we sample $N = 32$ outputs with a sampling temperature of 0.7. By default, we synthesize 2048 examples for fine-tuning, with context lengths randomly specified between $4K$ and $31K$ tokens. We conduct experiments using the Llama-3.1 models (Dubey et al., 2024) and Qwen-2.5 models (Yang et al., 2024a), with jina-embeddings-v3 serving as the sentence embedding model (Sturua et al., 2024). ORPO (Hong et al., 2024) is employed as the default fine-tuning method. More training details can be found in Appx. A.

4.2 Evaluation Setup

We conduct evaluations in long-context scenarios across a wide range of tasks. For single-document QA, we include Qasper (Dasigi et al., 2021) and MultiFieldQA-En (Bai et al., 2023) from the LongBench benchmark (Bai et al., 2023). For multi-document QA, we use HotpotQA (Yang et al., 2018), MuSiQue (Trivedi et al., 2022) and 2WikiMultihopQA (Ho et al., 2020), also from LongBench. Task statistics are presented in Tab. 3. We adopt plan-and-solve prompting for evaluation due to its strong performance (Tab. 1). Following previous research (Mallen et al., 2023; Asai et al., 2024; Yen et al., 2024b), we use substring exact match (SubEM) as the evaluation metric, measuring whether the output contains the golden answer.

4.3 Main Results

SEALONG Improves Various Models. We implement SEALONG on the leading open-source LLMs, including Qwen-2.5 models (Yang et al., 2024a) and Llama-3.1 models (Dubey et al., 2024). As illustrated in Tab. 2, SEALONG brings notable improvements: when implemented on Qwen-2.5-7B-Instruct, it closes the performance gap with

Model	Qasper	MultiFieldQA-En	HotpotQA	MuSiQue	2WikiMQA	Avg.
Llama-3.1-8B-Instruct	29.0	29.3	64.0	49.5	82.0	50.8
Supervised Fine-tuning						
+ TULU-V2-mix	26.5	27.3	49.5	27.5	54.0	37.0
+ WildChat	20.5	29.3	46.5	28.0	58.0	36.5
+ LongAlpaca	22.5	31.3	48.0	31.0	45.0	35.6
+ LongAlign	25.0	**36.7**	58.5	47.5	76.0	48.7
+ LongMIT	20.0	30.0	56.0	36.0	66.5	41.7
+ LongReward-SFT	22.0	28.7	58.0	52.0	76.5	47.4
+ GPT-4o-MuSiQue	21.5	31.3	64.0	**54.0**	83.5	50.9
+ SEALONG-SFT	**28.5**	30.7	**68.5**	50.5	**84.0**	**52.4**
Preference Optimization						
+ UltraFeedback	26.0	27.3	47.5	28.5	46.0	35.1
+ LongReward-Preference	26.5	**32.0**	63.5	52.0	80.5	50.9
+ SEALONG	**32.5**	31.3	**68.0**	**58.5**	**84.5**	**55.0**

Table 5: A comparison between SEALONG and previous datasets. The results are based on Llama-3.1-8B-Instruct finetuned on the corresponding dataset. To ensure fairness, $2K$ examples are randomly sampled from each dataset, with the exception of TULU-V2-mix, WildChat, and UltraFeedback, where the longest $2K$ examples are selected. The preference optimization strategy is ORPO (Hong et al., 2024).

Dataset	Supervision	Avg. Tokens
TULU-V2-mix (2023)	[1], [2], [3]	3,788
WildChat (2024a)	[2], [3]	32,230
LongAlpaca (2024b)	[1], [4]	9,160
LongAlign (2024)	[4]	16,881
LongMIT (2024c)	[5]	78,412
LongReward-SFT (2024b)	[6]	22,206
LongReward-Preference (2024b)	[6]	22,689
UltraFeedback (2023)	[3]	1,356
GPT-4o-MuSiQue	[7]	18,476
SEALONG	[8]	18,532

Table 6: Dataset statistics, including supervision source and average token count, measured with the Llama3.1-8B-Instruct tokenizer. Sources: [1] Human, [2] GPT-3.5-Turbo (OpenAI, 2022), [3] GPT-4 (Achiam et al., 2023), [4] Claude (Anthropic, 2023), [5] Qwen2-72B-Instruct (Yang et al., 2024a), [6] GLM-4 (GLM et al., 2024), [7] GPT-4o (Hurst et al., 2024), and [8] Self.

Qwen-2.5-14B-Instruct (51.8 vs. 52.2); when applied to Qwen-2.5-14B-Instruct, it even exceeds the performance of Qwen-2.5-32B-Instruct (54.7 vs. 53.1). Additionally, SEALONG yields an absolute improvement of 4.2 on Llama-3.1-8B-Instruct, outperforming GPT-4o (Hurst et al., 2024) (55.0 vs. 54.4). Although SEALONG utilizes MuSiQue for data synthesis, it achieves strong performance across other tasks as well, highlighting its generalization potential. One possible shortcut of SEALONG is producing more tokens, as the evaluation metric, SubEM, might favor outputs with more tokens, which are more likely to contain the golden answer. To explore this, we examine output token counts. As shown in Tab. 4, SEALONG has minimal effect on the number of output tokens.

SEALONG Competes with Previous Datasets. We compare SEALONG with several previous datasets, including short-context datasets such as TULU-V2-mix (Ivison et al., 2023), WildChat (Zhao et al., 2024a), UltraFeedback (Cui et al., 2023), as well as long-context datasets including LongAlpaca (Chen et al., 2024b), LongAlign (Bai et al., 2024), LongMIT (Chen et al., 2024c), and LongReward (Zhang et al., 2024b). Additionally, we utilize GPT-4o to synthesize data using the same question and long-context as SEALONG, creating a dataset we term GPT-4o-MuSiQue. Dataset statistics are presented in Tab. 6. To ensure fairness, $2K$ examples are randomly sampled from each dataset, with the exception of TULU-V2-mix, WildChat, and UltraFeedback, where the longest $2K$ examples are selected. As demonstrated in Tab. 5, most previous datasets negatively affect the performance of Llama-3.1-8B-Instruct, consistent with the observation of Gao et al. (2024). We hypothesize that this is because Llama-3.1-8B-Instruct already has strong long-context processing capabilities, and additional training on low-quality synthetic data could diminish its performance. However, we observe a performance improvement with SEALONG (50.8 to 55.0), indicating that self-improvement holds promise, which is particularlly promising as

current LLMs advance rapidly.

4.4 Analysis

Method	HotpotQA	MuSiQue	2WikiMQA
Greedy Search	64.0	49.5	82.0
Random	61.0	50.5	79.5
Reference-free Self-evaluation	64.0	51.5	83.0
Minimum Bayes Risk			
ROUGE	66.5	53.5	85.0
BERTScore	**67.5**	50.0	86.5
Reference-based Self-evaluation	63.5	51.5	84.5
Sentence Embedding	**67.5**	**56.0**	**88.0**

Table 7: Comparison of various scoring methods and greedy search. Each scoring method evaluates 16 outputs sampled from Llama-3.1-8B-Instruct. The results indicate the performance of the highest-scoring output for each method.

Scoring Methods. Effective scoring methods are critical for creating self-supervision signals (§3.1). We explore several approaches, including random selection, and reference-free self-evaluation, which prompts the LLM to assess its prediction in a separate turn based on the question and context. Additionally, we investigate various strategies for the utility metric $u(y, y^*)$ within Minimum Bayes Risk (§3.1), such as ROUGE (Lin, 2004), BERTScore (Zhang et al., 2019) and reference-based self-evaluation, which prompts the LLM to assess y using y^* as the reference. The detailed prompts for reference-free and reference-based self-evaluation are presented in Tab. 10 (Appx. B). For each question, we sample $N = 16$ outputs using a temperature of 0.7. Subsequently, we evaluate the highest-scoring output across different scoring methods and further compare these results to the performance of greedy search as a reference. As shown in Tab. 7, MBR-based methods outperform reference-free self-evaluation, even with simple N-gram-based ROUGE. We attribute this to the limited self-evaluation capabilities of current LLMs (Huang et al., 2024; Jiang et al., 2024), which might be more challenging in long-context scenarios. Incorporating more semantic information through sentence embeddings further improves MBR-based methods.

Number of Synthetic Examples. We analyze the impact of the number of training examples synthesized by SEALONG on long-context tasks. As shown in Fig. 3, SEALONG demonstrates strong data efficiency, achieving competitive performance with only $1K$ examples, after which additional examples provide limited benefit. This suggests that SEALONG is unlocking the inherent potential of LLMs for long-context reasoning rather than introducing a new skill that would require more data.

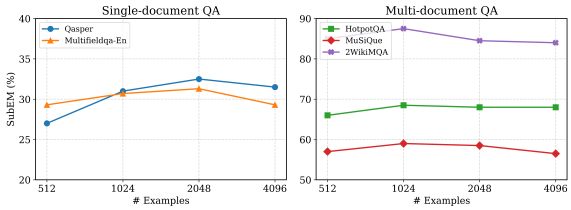

Figure 3: Long-context performance of SEALONG with varying numbers of synthetic training examples, evaluated based on Llama-3.1-8B-Instruct fine-tuned on the corresponding dataset.

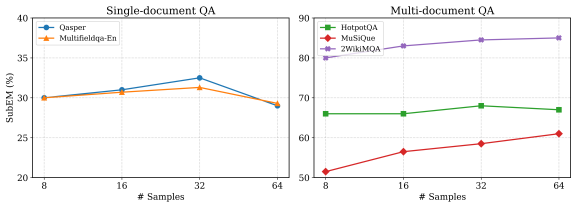

Figure 4: Long-context performance of SEALONG with varying numbers of samples per example during data synthesis, evaluated based on Llama-3.1-8B-Instruct fine-tuned on the corresponding dataset.

Number of Samples per Example. We continue to explore the effect of the number of samples, N, per example. As illustrated in Fig. 4, increasing N from 8 to 32 consistently improves performance, likely due to more accurate MBR estimation (§3.1). Beyond 32, except for MuSiQue, the performance improvement diminishes. This may indicate a fundamental limitation of our scoring method, which appears to struggle with selecting higher-quality outputs from larger output sets when $N > 32$ (see also in Tab. 1). We believe the scoring method is pivotal to self-improvement and will investigate this aspect further in future work.

Short-context Performance. Improving long-context reasoning should not compromise short-context performance. To investigate this further, we evaluate SEALONG on the Open LLM Leaderboard (Beeching et al., 2023), covering 6 tasks that represent diverse capabilities: MMLU (Hendrycks et al., 2021), GSM8K (Cobbe et al., 2021), ARC-challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2021) and TruthfulQA (Lin et al., 2022). As shown in Tab.

Model	Long-Context	Short-Context						
	Avg.	MMLU	GSM8K	ARC-Challenge	HellaSwag	Winogrande	TruthfulQA	Avg.
Qwen-2.5-7B-Instruct	49.0	74.2	82.4	67.1	81.5	74.7	64.7	74.1
+ SEALONG	**51.8**	74.1	83.2	66.5	81.3	74.4	64.8	74.1
Llama-3.1-8B-Instruct	50.8	68.3	77.7	60.2	80.1	77.4	54.1	69.6
+ SEALONG	**55.0**	68.4	77.8	60.3	79.9	77.3	53.8	69.6

Table 8: Evaluation results on short-context tasks from the Open LLM Leaderboard (Beeching et al., 2023), with the long-context average performance referenced from Tab.2. SEALONG demonstrates a marked improvement in long-context performance, with minimal impact on short-context performance.

8, while SEALONG achieves substantial improvements in long-context performance, it has minimal impact on short-context performance.

5 Related Work

Long-context Language Modeling. Numerous studies explore methods to extend the long-context processing abilities of LLMs. One line of research approaches addresses this challenge from a model-centered perspective, with some studies focusing on minimal modifications to existing LLMs, such as adjustments to position embeddings (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Zhu et al., 2024; Xiong et al., 2024) and refinements to the attention mechanism (Ding et al., 2023; Jin et al., 2024; An et al., 2024b,c). Additionally, some works propose novel architectures for efficient long-context processing (Wu et al., 2022; Bertsch et al., 2024; Wang et al., 2024d; Yen et al., 2024a; Lieber et al., 2024; Ye et al., 2024; Sun et al., 2024). Another line of research adopts a data-centric perspective, focusing on data engineering strategies. For example, Dubey et al. (2024); Lieber et al. (2024); Fu et al. (2024); Gao et al. (2024) continue pre-training models on long sequences, while An et al. (2024d); Bai et al. (2024); Zhang et al. (2024b); Chen et al. (2024c,b) leverage expert models or human annotations to create long-context data for fine-tuning. In contrast to these approaches, this work aims to facilitate the self-improvement of LLMs in long-context reasoning.

Self-improving. The self-improvement of LLMs has become a vital area of research as these models advance toward human intelligence. Research in this area follows two main approaches. The first approach investigates the self-reflection capabilities of LLMs, where models are prompted to assess and refine their own outputs (Ganguli et al., 2023; Madaan et al., 2024; Shinn et al., 2024; Xie et al., 2024; Gou et al., 2024; Chen et al., 2024a; Pan et al., 2024). However, the reliability of these self-refinement has been questioned in recent studies (Huang et al., 2024; Jiang et al., 2024). The second approach involves generating synthetic training data through the models themselves. This process typically involves generating multiple outputs for a given input, filtering out inaccurate results based on ground-truths, and using the remaining correct responses for model fine-tuning Zelikman et al. (2022); Hosseini et al. (2024); Pang et al. (2024); Wang et al. (2024c); Gulcehre et al. (2023); Zhang et al. (2024a). Additionally, Yuan et al. (2024) fine-tune LLMs to assign rewards to their own outputs using human preference data and facilitate continual improvement in instruction following. To reduce reliance on human annotations, some studies adopts consensus-based supervision, designating the output with the higher consensus across multiple outputs as better, with applications in areas such as arithmetic and logical reasoning (Huang et al., 2023; Prasad et al., 2024), machine translation (Finkelstein and Freitag, 2024; Wang et al., 2024a; Yang et al., 2024b), and instruction following (Wu et al., 2024). SEALONG first reveals the underestimated potential of LLMs in long-context reasoning and then leverages a consensus-based supervision strategy to enable LLMs to self-improve in long-context reasoning.

6 Conclusion

In this study, we investigate the potential of LLMs to self-improve in long-context reasoning and propose SEALONG for this purpose. This method achieves substantial improvements across multiple long-context reasoning tasks. We hope this research will open new avenues for self-improvement in long-context reasoning, which is vital for the sustained progress of LLMs, particularly as they advance toward surpassing human intelligence.

Limitations

We recognize that this work has several limitations that warrant further investigation.

Scoring Method. To establish self-supervision (§3.1), we score each output according to Minimum Bayesian Risk (MBR), which reflects consensus across multiple sampled outputs. However, a substantial performance gap remains between the highest MBR-scored output and the oracle sample (see Tab. 1 for details). Future research should explore more effective approaches for self-evaluation of outputs. One possible direction could involve examining the critic capabilities of LLMs in long-context scenarios (Lan et al., 2024b; Lin et al., 2024; Lan et al., 2024a).

Synthetic Data. Another limitation of this work is its reliance on MuSiQue (Trivedi et al., 2022) for synthetic data, which consists of multi-hop questions spanning multiple paragraphs. While this approach has enabled some progress, MuSiQue dose not cover all challenging question types, such as those requiring full-context reasoning, which remains a key limitation of current long-context LLMs (Karpinska et al., 2024; Wang et al., 2024b; Vodrahalli et al., 2024; Yen et al., 2024b). We advocate for future work to prioritize the creation of high-quality prompt sets, which are essential for the development of long-context LLMs.

Experimental Setup. Due to the computational limitations, we restrict the implementation of SEALONG to LLMs with up to 14B parameters, though its effectiveness at larger scales warrants further investigation. Likewise, the maximum sequence length is set to $32K$ tokens, whereas current leading LLMs support context lengths of up to $128K$ tokens or more. We leave the exploration of longer context lengths for future work.

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A Training Details

To support efficient fine-tuning for long-context scenarios, we implement sequence parallelization (Jacobs et al., 2023) with a parallel size of 8. Additionally, we utilize QLoRA (Dettmers et al., 2024) to reduce memory consumption during fine-tuning. The LoRA rank, alpha, and dropout are set to 128, 128, and 0.05, respectively, with all attention and feedforward linear layers designated as target modules. All models are fine-tuned for one epoch. The batch size, learning rate, and maximum sequence length are set to 8, $5e-5$, and $32K$, respectively. The β for ORPO is configured to 0.1. All experiments are conducted on a computing setup with $8 \times$ H100 GPUs.

B Prompts

We provide the prompts for various prompting strategies (§2.1) in Tab. 9, and the prompts for the reference-free and reference-based self-evaluation strategies (§4.4) in Tab. 10.

Strategy	Prompt
Default	{context} {input}
Direct Answer	{context} {input} Let's answer the question directly.
Think step-by-step (Kojima et al., 2022)	{context} {input} Let's think step by step.
Fact-and-reflection (Zhao et al., 2024b)	{context} {input} Let's first identify the relevant information from the long context and list it. Then, carry out step-by-step reasoning based on that information, and finally, provide the answer.
Plan-and-solve (Wang et al., 2023a)	{context} {input} Let's first understand the problem and devise a plan to solve it. Then, let's carry out the plan and solve the problem step-by-step.

Table 9: The prompts for various prompting strategies (§2.1), where {context} and {input} serve as placeholders for the long context and input query, respectively.

Strategy	Prompt
Reference-free Self-Evaluation	[Context] {context} [Question] {question} [Predicted Response] {prediction} Please evaluate the correctness of the predicted response based on the context and the question. Begin your evaluation by providing a brief explanation. Be as objective as possible. After giving your explanation, you must rate the response on a scale from 1 to 5, following this format exactly: "[[rating]]". For example, "Rating: [[3]]".
Reference-based Self-Evaluation	Here is a question along with two responses: one is the reference response, and the other is the predicted response. Please determine whether the two responses provide the same answer to the question. Respond with "True" or "False" directly. [Question] {question} [Reference Response] {reference} [Predicted Response] {prediction}

Table 10: The prompts for the reference-free and reference-based self-evaluation strategies (§4.4), where {question}, {reference}, {prediction}, and {context} serve as placeholders for their respective elements.